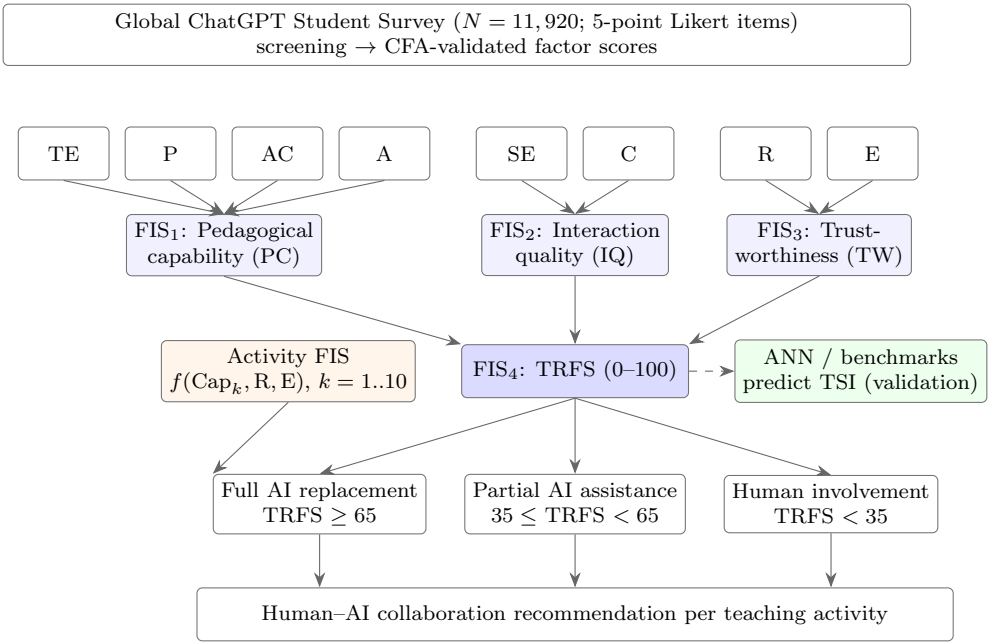

Global ChatGPT Student Survey ($N = 11,920$; 5-point Likert items)
screening → CFA-validated factor scores
TE
P
AC
A
SE
C
R
E
FIS$_1$: Pedagogical capability (PC)
FIS$_2$: Interaction quality (IQ)
FIS$_3$: Trust-worthiness (TW)
Activity FIS
$f(\mathrm{Cap}_k, \mathrm{R}, \mathrm{E})$, $k = 1..10$
FIS$_4$: TRFS (0–100)
ANN / benchmarks
predict TSI (validation)
Full AI replacement
TRFS ≥ 65
Partial AI assistance
$35 \leq$ TRFS < 65
Human involvement
TRFS < 35
Human–AI collaboration recommendation per teaching activity